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Amazon Product Review Analysis and Recommendations

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Declaration

This report is submitted in partial fulfilment of the requirement for the Master of Science in *Data Science and Analytics* at the University of Hertfordshire (UH).

It is my own work except where indicated in the report.

I did not use human participants in my MSc Project.

I hereby **give** permission for the report to be made available on the university website provided the source is acknowledged.

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Abstract

Online product reviews significantly impact consumer purchasing decisions. The sheer volume of data generated by these reviews can overwhelm consumers, making it difficult to assess them effectively. However, by employing machine learning algorithms to analyze the sentiments conveyed in these reviews, businesses can deliver more precise and personalized recommendations to their customers, enhancing the overall shopping experience.

The present research has commenced to detect user review sentiments along with recommending Amazon products. The datasets have been collected from Kaggle. The dataset comprises approximately 28,000 customer reviews on various Amazon products such as the Kindle, Fire TV Stick, and more. Text analytics have been performed to remove missing values, eliminate text noises, etc. Advanced preprocessing techniques such as tokenization, stemming, and lemmatization have been applied to ensure the data is clean and ready for analysis. TextBlob has been applied to extract polarity based on which the sentiment has been prepared based on the polarity values. Two feature extraction methods have been used namely TFIDF and Bag-of-Words methods with Unigram and Bigram to extract textual features from the cleaned review texts. The chosen models were applied to the data by sequential training and testing to get the sentiment prediction results.

The result of sentiment detection has been achieved optimally using Logistic Regression with Bag-of-Words and Unigram with an accuracy rate of 97.76%. Using this optimum model, Amazon's products have been recommended to users by taking necessary inputs.

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List of Abbreviations

LDA: Latent Dirichlet Allocation

XGBoost: Extreme Gradient Boosting

Word2vec: Word to Vector Model

TF-IDF: Term Frequency Inverse Document Frequency

SVM: Support Vector Machine

SPD: Sentiment Polarity Detection

SA: Sentiment Analysis

NN: Neural Networks

NLP: Natural Language Processing

NB: Naive Bayes

LSTM: Long Short-Term Memory

LSI: Large-Scale Integration

KNN: K Nearest Neighbours

BoW: Bag of Words

BERT: Bidirectional Encoder Representations from Transformers

API: Application Program Interface

ANOVA: Analysis of Variance

ABSA: Aspect-Based Sentiment Analysis

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Chapter-1 INTRODUCTION

1.1 OVERVIEW AND SCOPES OF PROJECT

1.1.1 Overview

In the era of e-commerce, online product reviews play a crucial role in influencing consumer purchasing decisions. With the vast amount of data generated by these reviews, it has become increasingly challenging for consumers to sift through and evaluate them effectively (Hu, et al., 2020). However, the application of machine learning algorithms has facilitated the analysis of sentiments expressed in these reviews, enabling businesses to provide more accurate and personalized recommendations to their customers.

Firstly, the importance of sentiment analysis in Amazon product reviews cannot be overstated. By leveraging natural language processing (NLP) and machine learning techniques, businesses can gain valuable insights into customer opinions and emotions regarding specific products. This allows for the identification of recurring patterns and trends, ultimately providing a comprehensive understanding of customer sentiments (Jain, et al., 2022). For instance, an analysis of product reviews may reveal common issues or strengths associated with a particular product, enabling Amazon to address concerns and enhance the overall quality of its offerings (Chehal, et al., 2022).

Moreover, the integration of machine learning models in the analysis of product reviews opens up opportunities for generating personalized recommendations. By identifying subtle nuances in customer sentiments, these algorithms can predict individual preferences with a higher degree of accuracy (Patidar & Patel, 2022). This, in turn, enables Amazon to offer tailored recommendations to its users based on their previous purchase history and expressed sentiments in reviews. The result is a more personalized and enjoyable shopping experience for customers, increasing the likelihood of them returning to Amazon for future purchases (Singh & Sarraf, 2020).

1.1.2 Scopes

The utilization of machine learning for sentiment analysis of Amazon product reviews presents a significant opportunity for enhancing customer satisfaction and driving sales. By accurately understanding and interpreting the sentiments expressed in reviews, Amazon can improve the relevance and precision of its product recommendations, ultimately leading to a more

personalized and satisfying shopping experience for its customers (Zarzour, et al., 2021). Therefore, the integration of machine learning algorithms in sentiment analysis has the potential to revolutionize the way that Amazon and other e-commerce platforms interact with their customers, ultimately benefitting both consumers and businesses.

1.2 CHALLENGES

The exploration of recommending e-commerce or online products to consumers encompasses a set of challenges that could potentially complicate the execution of the research artefact. Addressing and mitigating these issues is crucial for the successful completion of the research. This necessitates a proactive approach to identify, understand, and resolve potential obstacles, thereby ensuring the integrity and effectiveness of the research outcomes.

In the context of e-commerce, the significance of tailored product recommendations cannot be overstated, as they play a pivotal role in enhancing the customer shopping experience and driving sales. As highlighted by recent findings, a substantial percentage of online shoppers show a preference for purchasing from platforms that offer personalized recommendations (Chehal, et al., 2022). This underscores the critical need for e-commerce small and medium-sized businesses (SMBs) to leverage accurate and effective recommendation systems to meet consumer demands and stay competitive.

Additionally, the challenges of cross-border e-commerce expansion and the necessity of efficient processes and technology alignment are pivotal considerations for businesses aiming to optimize their e-commerce strategies (Singh & Sarraf, 2020). By focusing on overcoming these common issues through strategic improvements in product offerings and customer experience, businesses can achieve significant advancements in operational efficiency and revenue growth.

The possible challenges which may be seen at the time of the project are mentioned below:

1. The study will be undertaken utilizing review data, sourced in the form of textual information. This particular type of data is not inherently compatible with processing algorithms and requires preliminary handling.
2. Text preprocessing presents a notable challenge within text-centric research contexts due to the plethora of available methodologies, such as Term Frequency-Inverse Document Frequency (TF-IDF), Bag of Words, and Word to Vector, among others.

Selecting an appropriate method necessitates thorough investigation, which will be meticulously conducted as part of this research.

3. Furthermore, the review data may not explicitly convey the sentiment of the user, a critical factor in gauging demand. In instances where sentiment is not directly discernible, a precise sentiment extraction process will be employed to ensure accurate analysis.

1.3 PROJECT AIM

The project aims to analyze the user review sentiments of Amazon products using natural language processing and machine learning to recommend suitable products to the users according to their demand.

1.4 RESEARCH QUESTIONS

- RQ-1.** What are the most effective text vectorization techniques for sentiment analysis of product reviews in online marketplaces, and how do they impact the accuracy of sentiment detection?
- RQ-2.** Which machine learning algorithms are most suitable for detecting sentiment in product reviews, and how do their performance metrics compare regarding accuracy, precision, recall, and computational efficiency?
- RQ-3.** How can user preferences and historical data be incorporated into the recommendation system to improve the accuracy and relevance of product recommendations in online marketplaces?

1.5 PROJECT OBJECTIVES

The objectives to be done in the project are stated below with their relevant outcomes:

- RO-1** Develop and implement advanced natural language processing techniques to extract sentiment from Amazon product reviews, evaluating various machine learning classifiers to identify the most effective models.

Outcome: By combining sentiment extraction with classifier evaluation, the aim is to establish a robust framework for accurately analyzing customer opinions, facilitating informed decision-making for both consumers and sellers.

- RO-2** Employ machine learning algorithms to classify sentiment in Amazon product reviews effectively, integrating the results into a recommendation system.

Outcome: Integrating sentiment analysis into the recommendation system enhances the personalized shopping experience for customers, leading to increased user engagement and satisfaction while automating sentiment classification for scalable implementation.

RO-3 Validate the sentiment-based recommendation system through real-world product recommendations, including a comparative analysis with previous research to assess advancements and the effectiveness of the proposed approach.

Outcome: Validating the recommendation system in real-world scenarios not only ensures its practical utility but also provides insights into potential enhancements. Comparing the accuracy of the optimized model with that achieved in previous research within the same field contributes to understanding the advancements made and the effectiveness of the proposed approach, guiding future research efforts and enhancing the relevance of sentiment analysis and recommendation systems in the dynamic e-commerce landscape.

Chapter-2 PROJECT MANAGEMENT

2.1 PROJECT AND RESEARCH ISSUES

2.1.1 Social Issues

The application of ML in analyzing product reviews on platforms like Amazon has profoundly impacted social dynamics. On one hand, it democratizes consumer power, enabling individual voices to influence product quality and brand reputation significantly. Conversely, it raises concerns regarding privacy and data manipulation. The aggregation and analysis of customer sentiments could inadvertently reveal personal preferences and sensitive information, leading to potential misuse (Zuboff, 2019).

2.1.2 Legal Issues

Legally, the use of sentiment analytics treads a fine line between innovation and infringement. The General Data Protection Regulation (GDPR) in the European Union and similar laws in other jurisdictions place strict limits on the collection, processing, and use of personal data (European Commission, 2018). Amazon's global operations necessitate adherence to these laws, requiring transparent consent from users for data usage and providing them with options to opt-out. Moreover, the accuracy of ML algorithms in sentiment analysis and the potential for biased outcomes pose legal challenges, potentially leading to defamation or misrepresentation.

2.1.3 Ethical Issues

Ethically, the deployment of ML for sentiment analytics on Amazon reviews is a double-edged sword. While it can enhance user experience and provide valuable insights for product improvement, it also raises questions about algorithmic bias and fairness. ML algorithms are only as unbiased as the data they are trained on; any inherent biases in customer reviews could be amplified, leading to unfair product ratings and reputational damage for sellers (EPRS, 2020). Additionally, the ethical use of customer data for training these algorithms without explicit consent is a contentious issue.

2.1.4 Professional Issues

From a professional standpoint, the implementation of sentiment analytics using ML requires a high level of expertise in data science and machine learning. This necessitates ongoing

education and training for professionals in the field to stay abreast of the latest developments and ethical guidelines (Baker, 2021). Moreover, professionals are tasked with ensuring the accuracy, reliability, and transparency of the algorithms they develop, a significant challenge given the dynamic nature of language and sentiment.

2.1.5 Organizational Issues

For Amazon and similar e-commerce entities, integrating ML-based sentiment analytics into their operational framework presents several organizational challenges. These include the allocation of resources for the development and maintenance of these systems, the management of vast amounts of data, and ensuring the privacy and security of customer information (Ducato, 2020). Additionally, there is the challenge of integrating these insights into strategic decision-making processes, requiring a cultural shift towards data-driven decision-making.

2.1.6 Business Issues

On the business front, the use of sentiment analytics offers a competitive edge, enabling companies like Amazon to tailor their offerings and improve customer satisfaction. However, this also raises the issue of dependency on technology, where businesses may become overly reliant on algorithms for decision-making, potentially overlooking human intuition and experience (The Data Protection Act, 2019). Furthermore, the competitive advantage gained through advanced ML algorithms could exacerbate market dominance concerns, raising antitrust issues in some jurisdictions.

2.2 PROJECT PLANNING

The Gantt chart for the project has been presented below - *For the duration of my second-year semesters B and C, totalling 180 days, I am dedicating 600 hours to my project, aligning with the standard time allocation for all participants.)*

TASK COMPLETION			DAYS					
Task Name	Subtasks	Status	Days 1-40	Days 41 - 120	Days 120-145	Days 146-160	Days 161-165	Days 166-180
Introduction	Overview	Complete						
	Background Study	Complete						
	Aim and Objectives	Complete						
	Research Questions	Complete						
Literature Review	Finalize	Complete						
	Draft	Complete						
	Complete	Complete						
	Comparison	Complete						
Methodology	Finalize	Complete						
	Framing Approach for Research	Complete						
	Steps of Methodology for Research	Complete						
	Tool Selection	Complete						
Result and Analysis	Data Selection	Complete						
	Algorithm Selection and Determining Evaluation Matrices	Complete						
	Selection of Feature Extraction Methods (TFIDF & Count Vect with Two n-grams)	Complete						
	Documentation	Complete						
Discussion and Conclusion	Analysis of Product Reviews	Complete						
	Text Preprocessing and Analytics (Cleaning etc.)	Complete						
	Sentiment Extraction	Complete						
	Feature Extraction	Complete						
Finalize Document	Classifying Sentiment of Review using Chosen Models	Complete						
	Selection of most-Effective Model and Test it	Complete						
	Recommendation of Products	Complete						
	Documentation	Complete						
Finalize Document	Addressing Research Questions	Complete						
	Research Limitations and Future Scopes	Complete						
	Advantages of Research	Complete						
	Finalize	Complete						
Finalize Document	Submission	Complete						

Figure 1 Planning and Gantt Chart

Chapter-3 LITERATURE REVIEW

3.1 SENTIMENT ANALYTICS ON USER REVIEW FOR E-COMMERCE PRODUCTS

Sentiment analysis using the Bengali language has recently seen a surge in scholarly interest (Akter, et al., 2021). Opinion extraction along with trend prediction relies heavily on this method, which can extract sentiment through the ever-present text reviews seen online. This research presents an ML model that uses methods including XGBoost along with Random Forest as well as Logistic Regression with SVMs as well as KNN to forecast the tone of Bangla text critiques. Out of all of them, KNN outperforms the others with a precision of 96.25% along with a well-balanced f1-score of around 0.96.

(Liu, et al., 2018) presented the application of FL-SVM techniques to improve the accuracy of sentiment categorization in online evaluations of product quality. The way Net is used to construct a dictionary of fuzzy sentiment via fuzzy theory for determining the polarity of emotions. A classifier for assessing sentiment tendencies is constructed using SVM. Experimental comparisons show that FL-SVM achieves a 1% to 3% improvement in accuracy throughout different datasets compared to NB, kNN, as well as conventional SVM, confirming the effectiveness and robustness of the suggested methods.

As stated by (Yao, et al., 2022), An increasing number of people are taking to the Internet, especially on e-commerce sites, to voice their thoughts and share their ideas in recent years. This research delves into sentiment analysis employing data from several sources, with a specific emphasis on evaluations of product outliers. It uses a statistics-based detection of outliers as well as a correction approach i.e., SODCM to look at the difference among sentiment as well as star ratings, particularly in 3-star feedback. Investigating how SODCM behaves while labelling 3-star evaluations as either favourable or adverse, the research delves into the polarity assignment process.

According to (Patel & Patel, 2018), There is an overwhelming quantity of information available to customers due to the explosion of goods and services brought about by the advent of online marketing through Internet services. The presence of fraudulent reviews is a major obstacle in this ecosystem because user-generated ratings significantly impact decision-making. This literature review delves into tactics to tackle this problem by exploring several data mining

approaches that may identify and reduce the effect of phoney reviews. Methods examined include supervised, unsupervised, as well as semi-supervised methods.

Online product evaluations have a significant impact on customer behaviour and (Shah, et al., 2021) delve into the role of sentiment analysis within the modern technological world. User feedback serves as crucial for purchasing choices, as well as major e-commerce sites including Amazon, Myntra, as well as Flipkart make it easy for customers to provide it. Sentiment analysis uses ML algorithms to classify reviews as either favourable, neutral, as well as negative. By comparing Random Forest along with Naive Bayes, as well as logistic regression using information gathered through Kaggle's Amazon item evaluations, find that Random Forest provides the best as well as most accurate method.

3.2 FEATURE ENGINEERING FROM TEXT DATA

Classifying text efficiently is becoming more important due to the exponential rise of digital documents containing text (Shah & Patel, 2016). Nevertheless, efficient classification is hindered by the difficulty of dealing with feature spaces with high dimensions. Approaches for the extraction, as well as selection of features, arise as critical answers; they improve classification performance by lowering dimensionality as well as removing superfluous features. Improving text document classification is the focus of this paper, which delves into several methods for characteristic selection/extraction including classifiers.

In order to classify brief phrases along with sentences employing neural networks, (Dzisevič, 2019) investigated three different approaches to text extraction of characteristics. After comparing TF-IDF using LSA as well as LDA, as well as with TF-IDF itself, the results show that TF-IDF achieves the best accuracy with bigger datasets. Also, with smaller datasets, the accuracy is equivalent when TF-IDF and LSA are combined. The usefulness of TF-IDF towards collecting text characteristics and improving classifier performance throughout different dataset sizes is highlighted by these findings.

As reported by (Du & Hu, 2022), customer support at the Power Supply Bureau is overloaded due to the recent influx of complaints, which has caused delays in problem resolution. In order to tackle this, provide a technique for detecting text similarity in power support work orders that is centred on TFIDF. Employs web scraping for gathering texts from work orders, preprocessing to separate Chinese text, as well as TFIDF for gathering keywords. With better

F1 scores alongside reduced complexity, the results show that the new technique is more efficient than the old one.

The relevance of text clustering in NLP and other areas of pattern recognition and machine learning is discussed by (Wang, et al., 2012). Using the Naïve Bayes classification algorithm, the article suggests an improvement to the traditional TFIDF technique. Feature word choice, as well as classification, are enhanced by iterative refinement, leading to increased efficiency along accuracy. The results of the experiments show that the method is effective for improving classification along with feature selection.

(Hamisu & Mansour, 2020) delved into Advance Fee Fraud i.e., AFF, a kind of Internet fraud that is very common and has a substantial impact on people's finances throughout the world. The paper suggests a Fraud Email Classification which employs NLP, more especially the BoW model. For training as well as verifying the classifier, need a dataset to differentiate between fraudulent as well as non-fraudulent emails; use Nigeria as an example. The approach's efficacy in fighting AFF is emphasised by the experimental findings, which show good performance.

According to (Li, et al., 2017), With the rise of e-commerce platforms, consumers are more dependent on reviews to help them make purchases, which has led to an increase in fraudulent reviews designed to boost merchants' profits. There is a need to distinguish between misleading and honest reviews because of these fake ones, which are usually written by paid people without real product expertise. The research examines the effectiveness of several classifiers by using word embedding to feed contextual semantic evaluation and by combining TF-IDF weighting. Results demonstrate the method's resilience; achieved an impressive 93.6% accuracy in detecting misleading reviews, and confirmed findings among English as well as Chinese datasets.

The disparity among e-commerce platforms' rating systems, as well as textual reviews, is brought to light throughout by (Singh & Sarraf, 2020). There are a lot of ways to analyse emotion, but star ratings aren't always reliable. A lot of things get good ratings even when they get bad evaluations. This problem is made worse by the fact that there is no universally accepted ranking system. To get around this, the suggested solution would provide customers with a simple yes/no answer about the product's quality rather than making them go through all the reviews.

Data creation has surged due to the fast expansion of information technology, especially from internet sources like product and service evaluations as stated by (Imaduddin, et al., 2019). Effective data processing systems are a must for this. One method that has proven useful in assessing service quality, especially in the hospitality sector, is sentiment analysis, a branch of NLP. This research uses Traveloka evaluations of hotels to examine the performance of several word incorporation techniques using the LSTM algorithm. These methods have been CBOW along with skip-gram or doc2vec, as well as a glove. With an accuracy of 95.52%, the glove is clearly the best word embedding approach for hotel feedback data, according to the results.

3.3 SENTIMENT DETECTION ON AMAZON PRODUCT REVIEWS

3.3.1 Sentiment Analytics Using Machine Learning

According to (Zhao, et al., 2021), as internet technology has advanced, online purchasing has become commonplace. Customer happiness on e-commerce sites is boosted by using sentiment analysis upon review text. Sentiment polarity prediction is still not easy, nevertheless, because of complicated reasoning and varying sequence lengths. This work presents LSIBA-ENN, an ML method that has been optimised for SA. Gathering information, extraction of features, sentiment categorization, and data collecting are all part of the suggested approach. LSIBA-ENN achieves greater recall rates than current algorithms, showing that it is useful for sentiment categorization.

(AlQahtani, 2021) researched the rapidly expanding field of Internet shopping and how it has changed the way consumers read reviews and make purchases. Using a dataset of Amazon reviews, the study investigates sentiment categorization using several ML techniques, such as Bag-of-words along with Tf-Idf, as well as GloVe. Several algorithms are used, including BERT and Naïve Bayes along with Bidirectional LSTM, as well as Logistic Regression, along with measures including preciseness as well as accuracy are used for assessment. After starting with multiclass classification, research moves forward to binary categorization and finally finds the best model.

As stated by (Sultan, 2022), businesses may benefit greatly from the plethora of product reviews that have proliferated with the rise of Internet shopping. Analysing customer sentiment via phone reviews is the topic of this literature study. By classifying views as favourable, negative, as well as neutral, here sentiment evaluation helps businesses improve their products

and customers make better decisions. Accurate categorization relies on preprocessing procedures, such as eliminating punctuation including stop words, as well as evaluation using a variety of techniques of machine inspection.

To classify internet reviews as favourable, neutral, as well as negative, (Rathor, et al., 2018) investigated the analysis of sentiment employing NLP, specifically looking at the effectiveness of SVM, Naive Bayes, as well as Maximum Entropy. Trained ML classifiers employing unigrams as well as weighted unigrams, as well as used reviews acquired via the Amazon API. The results show that when using weighted unigrams, SVM achieves much better accuracy. Companies that track public comments and customers who are looking for product evaluations may benefit from this study.

The rapidly expanding area of Sentiment Analysis under Information Extraction studies online discourse on a wide range of topics (Sarulkar, et al., 2023). One such example is Amazon, where customers can easily help each other out by leaving ratings and reviews on things. Both academics and businesses are interested in SA because of the gap it fills between NLP and ML. Improvements to current techniques, especially those using supervised and mixed ML, have been the subject of recent research. Among the suggested enhancements is a reduction of the input information preprocessing covariance.

As per (Huiliang, et al., 2021), due to the proliferation of high-speed Internet, buying goods and services online has become the norm. It is of the utmost importance to increase consumer satisfaction by analysing the sentiment of online reviews. An improved ML method for SA, LSIBA-ENN, is presented in this paper. Gathering data, cleaning it up, extracting features, and classifying sentiment are all part of it. With an improved recall score of 87.79% as opposed to ENN's i.e., 83.55 using the LTF-MICF scheme, LSIBA-ENN surpasses all previous algorithms. When it comes to analysing reviews of products on the internet, LSIBA-ENN shows great promise because of its sentiment categorization capabilities.

Customers now have unparalleled access to items without ever leaving the comfort of their homes, thanks to the rise of e-commerce made possible by the digitization of the contemporary world (Haque, et al., 2018). The importance of evaluations of goods has skyrocketed due to the growing dependence on internet buying. Using models to sift through and categorise mountains of reviews is a time-saving strategy in this machine-learning age. Supervised learning used to large Amazon collections improved review polarisation accuracy feasible level.

As stated by (Monsoor, et al., 2023), in the world of online shopping, customer evaluations are very important for building a company's reputation. A vital component of NLP, sentiment analysis has been emphasised in this research for evaluating review tone. Multiple classifiers including Multinomial Naïve Bayesian along with SVM, or Maximum Entropy, as well as Logistic Regression are used to categorise feedback from Amazon in the investigation's technique. The work accomplishes remarkable accuracy in emotion polarity classification by using ML on a large dataset.

(Rao & Sindhu, 2021) have shown that people are increasingly sharing their thoughts and feelings on social media, which calls for an examination of these feelings. From direct to snarky, comments often convey subtle feelings. Using sentiment analysis tools, these intentions may be deciphered. Data preparation, which includes operations like stemming, polarity identification, and tokenization, follows dataset selection through sources including Amazon. The n-gram as well as term frequency methods of extraction are used. Use random forests along with K-nearest neighbours, as well as SVMs for classification, leveraging accuracy as the primary measure of success.

Through an examination of online product evaluations, (Kumar, et al., 2021) delved into sentiment analysis in the context of the contemporary technological age. Online marketplaces such as Amazon, Flipkart, as well as Myntra, rely heavily on customer reviews to inform product suggestions and ultimately, sales. A dataset acquired from Kaggle is used for sentiment categorization using machine learning methods, namely Logistic Regression along with Naive Bayes, as well as Random Forest. According to the results, Random Forest has become the best algorithm to use for this specific assignment.

Examining the relevance of consumer sentiment for businesses and producers, (Kumar, et al., 2022) focused on the setting of immediate product and service reviews. But with so much data, it would be unrealistic to carefully analyse every single online message. Sentiment analysis is therefore a powerful method for processing data efficiently and widely. Utilising Python as well as ML Algorithms, this research delves into the practical use of sentiment analysis in the corporate world, examining its strengths and weaknesses via the lens of the Amazon Alexa evaluation dataset.

3.3.2 Sentiment Analytics Using Deep Learning

According to (Sangeetha & Kumaran, 2023), consumer ratings are crucial when it comes to gauging company performance and influencing customer decisions. Methods based on

Sentiment Analysis find sentiments in massive review datasets. Feature selection is a great tool for zeroing down on the specific product qualities that matter. This research presents a PCCHHO-RNN-LSTM technique that uses Harris Hawks Optimisation as well as the coefficients of Pearson correlation to improve the accuracy of sentiment categorization. By using Amazon.com information, the approach significantly enhances performance, claiming a 95.8% correctness rate, 95.4% accuracy rate, 95.6% recall rate, as well as 95.2% F-measure.

Online shopping has skyrocketed, and with it, the number of product reviews (Mehul, et al., 2023). This is all down to the exponential rise of e-commerce. Opinions like this have a major impact on what people buy. One of the most important methods for understanding and classifying textual data is sentiment polarity analysis. The objective of the research is to sort evaluations written by customers into good and bad categories. Use a holistic approach by using tools including Tf-Idf as well as Tokenizer for preprocessing, the extraction of features, training, as well as classification. Considering an impressive 86 per cent precision with Amazon reviews as well as an even better 85% precision rate for reviews of Yelp, examination, which includes SVM Linear along with RBF or Sigmoid kernel as well as LSTM models, shows promising accuracy levels.

In this age of rapidly developing technologies and ever-increasing online sales, (Ben, et al., 2023) delved into the critical function of customer evaluations as indicators of product quality. Using machine learning as well as hybrid learning techniques that include several deep neural networks, it places an emphasis on sentiment polarity identification, which is essential for wise decision-making. This study uses TF-IDF vectorization for assessing many models for identifying the polarity for sentiment within the Amazon information set, including ML, DL, as well as hybrid approaches. The integrated deep learning method outperforms more conventional ML techniques, according to the results.

An increase of 43% in online sales during the COVID-19 epidemic is indicative of the growing importance of this sector (Sumyta, et al., 2023). Amazon, a well-known online retailer, is well-known for its extensive product catalogue. While reading reviews is essential before making a purchase, it may take a lot of time. The BERT, RoBERTa, as well as XLNet models, are used to analyse the sentiment of Amazon reviews in this research. Its goal is to improve customer decision-making by categorising attitudes and identifying features that differentiate better items.

(Alzahrani, et al., 2022) looked at the problem of misleading reviews as well as how important online reviews are for e-commerce making choices. In order to improve market tactics, it presents a new method that uses opinion mining. Amazon reviews of different goods, the research achieved 94% using LSTM as well as 91% accuracy using CNN-LSTM algorithms using sentiment analysis. These results demonstrate that deep learning approaches are capable of accurately gauging customer sentiment towards items.

Amazon product sentiment analysis is the focus of this research done by (Alroobaea, 2022). Opinions have a significant impact on people's actions in many social and professional settings. Classifying reviews from three separate Amazon datasets, the research achieves 85% accuracy using RNN models, accordingly. The results show that RNN is successful during sentiments of predicting author, and they are comparable to leading models.

3.4 PRODUCT RECOMMENDATION SYSTEM

3.4.1 Using Machine Learning and NLP

With more and more people buying online as well as participating in digital globalisation, effective product suggestion systems are needed (Sharma, et al., 2023). Customer happiness and corporate earnings are being hurt by the absence of personalised advice in current approaches. Using the database of Amazon Apparel, this research suggests a new method that makes use of NLP and CNN technology. The system improves the user experience and increases the chance of purchase by 35% by predicting related goods based on product names and picture attributes obtained through VGG-16 architecture.

The overwhelming amount of items has been caused by the fast proliferation of e-commerce platforms (Dwivedi, et al., 2020). To help customers in traversing these massive resources, algorithms for recommendation have been implemented. In order to improve the user experience and increase sales, this article investigates the different recommendation methods used by firms such as Amazon. It explores several assessment methodologies and discusses problems with such systems. Leveraging data from Amazon's electronics store, it also delves into the creation and assessment of recommender engines that rely on popularity as well as collaborative filtering.

As reported by (Marchand & Marx, 2020), automated item recommender systems play an important part in online shopping, as this literature review shows, particularly in increasing

cross-selling as well as client loyalty. Errors may still happen in such systems, even with all the improvements, along that can damage the confidence and contentment of customers. To increase accuracy as well as provide clear explanations, the paper suggests a unique technique that combines content-based as well as shared filtering. Recommender systems' operational, as well as ethical issues, are addressed by the results, which show greater predicted accuracy as well as the capacity to give actionable answers.

Based on a sociological theory that states that businesses promote preference stability in order to boost predictability as well as profits, (Rystrøm, 2023) investigated how recommender systems might influence user preferences. Using Barrier-to-Exit measurements, the study analyses Amazon Book Reviews spanning 1998 towards 2018 to measure the difficulty of changing preferences. As time goes on, the results show that Barrier-to-Exit increases significantly, which means that preference inertia becomes worse. The work highlights the need for sophisticated methods to address the hazards of preference manipulation, notwithstanding the methodological hurdles.

From digital assistants towards real-time internet recommendations, (Necula & Păvăloaia, 2023) covered it all in terms of the deep connection among electronic commerce as well as recommendation systems. For different jobs and kinds of data, different algorithms as well as technologies are used. In particular, AI makes use of ML and deep learning methods, with textual data being subjected to predictive modelling and visual data for image processing. This analysis takes a methodical look at how e-commerce suggestion systems use AI, finding ways that AI may work in tandem with blockchain, VR and AR to improve the user experience. (Pawlick, et al., 2021) explored the crucial function of recommendation systems in cybersecurity. It gives a comprehensive rundown of the many kinds of recommender systems, including their benefits, downsides, possible uses, alongside associated security issues. Also included are current solutions as well as creative future possibilities for recommender system utilisation within cybersecurity, as well as a summary of the present landscape. Addressing a vacuum in the literature, the paper's thorough review unites numerous recommender kinds that have been mostly ignored.

As stated by (Chen, et al., 2017), recommender systems that use machine learning techniques have recently seen a rise in attention. Scalability, precision, as well as durable systems, are essential for successful algorithms in light of the profusion of explicit and implicit user preference indications. An improved latent factor approach that uses matrix factorization, the

ALS technique is very scalable but has very good prediction accuracy. The combination of Apache Spark which is a distributed system for processing massive amounts of information, with AWS provides a scalable as well as highly available approach. The research demonstrates impressive scalability as well as accuracy regarding product recommendations employing Apache Spark over AWS EMR.

3.4.2 Using Deep Learning

A growing body of research done by (Li, et al., 2023) in fields as varied as e-commerce as well as the entertainment industry highlights the critical function of systems of recommendations in increasing both consumer happiness and revenue. Methods based on deep learning have significantly enhanced the precision and efficacy of the system. However current systematic reviews are either too restricted in their emphasis on certain areas or fail to cover all the bases, leaving out innovative methods. This study aims to fill these gaps by providing a thorough evaluation of deep learning approaches utilised by recommendation systems, specifically for word categorization. It sheds light on popular methods such as neural networks with graphs as well as highlights potential applications such as social connections as well as e-commerce. With its helpful insights, the present research addresses a critical gap in the existing literature and paves the way for further research in the area.

(Zhang, et al., 2023) delved into the growing field of e-commerce along with its relationship with ML as well as DL methods. A variety of neural networks, decision trees, and support vector machines are among the methods examined. Sentiment analysis, engines for recommendation, as well as detecting fraud are some of the main subjects discussed. Emphasis is placed on difficulties such as unbalanced data along with interpretability. This summary highlights the need for continuous research in order to successfully navigate the ever-changing e-commerce environment.

According to (Paranjape, et al., 2021), modern business relies heavily on systems of recommendation, which personalise recommendations according to consumer preferences. This research supports a collaborative filtering approach that uses ML methods to provide product suggestions. When dealing with sparse data, using singular value decomposition may help. System correctness is guaranteed by the examination of Root Mean Square Error. The importance of personalised suggestions in improving user experience and generating sales is shown by this technique, which is seen in platforms including Amazon and Flipkart.

As per (Rahman, et al., 2023), online shopping along with product reviews have both seen significant growth in recent years, coinciding with a meteoric boom in e-commerce. Consumers' choices are greatly impacted by these reviews. Determining consumer sentiment relies heavily on the polarisation of sentiment analysis, which comprises text interpretation along with categorization. The goal is to classify reviews as either favourable or adverse using methods including feature extraction, preprocessing, as well as ML models involving SVM as well as LSTM. Promising findings are shown by evaluation measures including recall, precision along with accuracy, as well as f1-score. LSTM model achieved 85% accuracy with Yelp reviews as well as 86% accuracy with Amazon reviews.

Recommender systems are crucial to the viability of digital commerce due to the exponential expansion of online information (Zarzour, et al., 2020). On the other hand, most systems put an emphasis on accurate recommendations rather than explainability. Research fills this need by presenting a new deep learning structure to generate explainable suggestions based on lengthy short-term recollections. Users may confidently make educated selections with the help of clear and concise explanations along with suggestions. The method successfully improves suggestion variety along with accuracy, as shown by extensive testing on Amazon datasets.

3.5 GAP ANALYSIS

With the review of existing research, some gaps have been identified regarding the methods applied for text analytics and recommendation processes which will be fulfilled in this present research.

Table 1 Identified Gaps

Researchers	Sentiment Analytics	Product Recommendation	Feature Extraction Method	Multiple Feature Extraction	Method	Application of Machine Learning	Application of Deep Learning
Mehul et al. (2023)	✓	✗	TFIDF	✗	SVM	✓	✗
Alroobaea (2022)	✓	✓	Word2Vec	✗	RNN	✗	✓
Rahman et al. (2023)	✓	✓	BOW	✗	LSTM	✗	✓
Huiliang et al. (2021)	✓	✗	BOW	✗	LSIBA-ENN	✗	✓
Alzahrani et al. (2022)	✓	✗	Word2Vec	✗	CNN-LSTM	✗	✓
Sangeetha & Kumaran (2023)	✓	✗	BOW	✗	LSTM	✗	✓
Imaduddin et al. (2019)	✓	✗	CBOW	✗	LSTM	✗	✓
Akter et al. (2021)	✓	✗	TFIDF	✗	KNN	✓	✗

Chapter-4 METHODOLOGY

4.1 DATA SELECTION

4.1.1 Data Description

There is a multitude of online marketplaces available worldwide that customers utilize to purchase goods and products. However, not all online marketplaces meet the highest standards. Furthermore, within a single marketplace, not all products may be satisfactory.

Amazon, being a prominent global online marketplace, has been selected as the source of user reviews for online products. The dataset comprises approximately 28,000 customer reviews on various Amazon products such as the Kindle, Fire TV Stick, and more, sourced from Datafiniti's Product (Datafiniti, 2019). The partial data is shown below:

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Figure 2 Amazon Product Data

4.1.2 Feature Description

The selected dataset has 24 features and the description of the data features is as follows:

Table 2 Feature Description

Feature	Description	Feature	Description
name	Product Name	reviews.doRecommend	Significance whether the product can be recommended
dateUpdated	Date of Review Updating	reviews.didPurchase	Significance whether the product has been purchased
manufacturer	Manufacturer of the product	reviews.username	Username of the reviewer
brand	Brand name	reviews.numHelpful	Significance whether the review is helpful
asins	Assigned product	reviews.id	ID of the reviewer
manufacturerNumber	Number of the Manufacturer of the product	sourceURLs	URL to access the user profile
primaryCategories	Primary Category of the enlisted products	reviews.sourceURLs	URL to access the review text
keys	Product Key	reviews.title	Title of the review
imageURLs	URL to access the image of the product	reviews.text	Text of the Review
categories	Category of the enlisted products	reviews.rating	Product rating
dateAdded	Date of adding the product	reviews.dateSeen	Date when the review has been seen
id	ID of the review on products	reviews.date	Date of review

4.2 PROPOSED METHODOLOGY

The proposed methodology for sentiment analytics and recommending Amazon products, based on which the artefact will be prepared, is presented below:

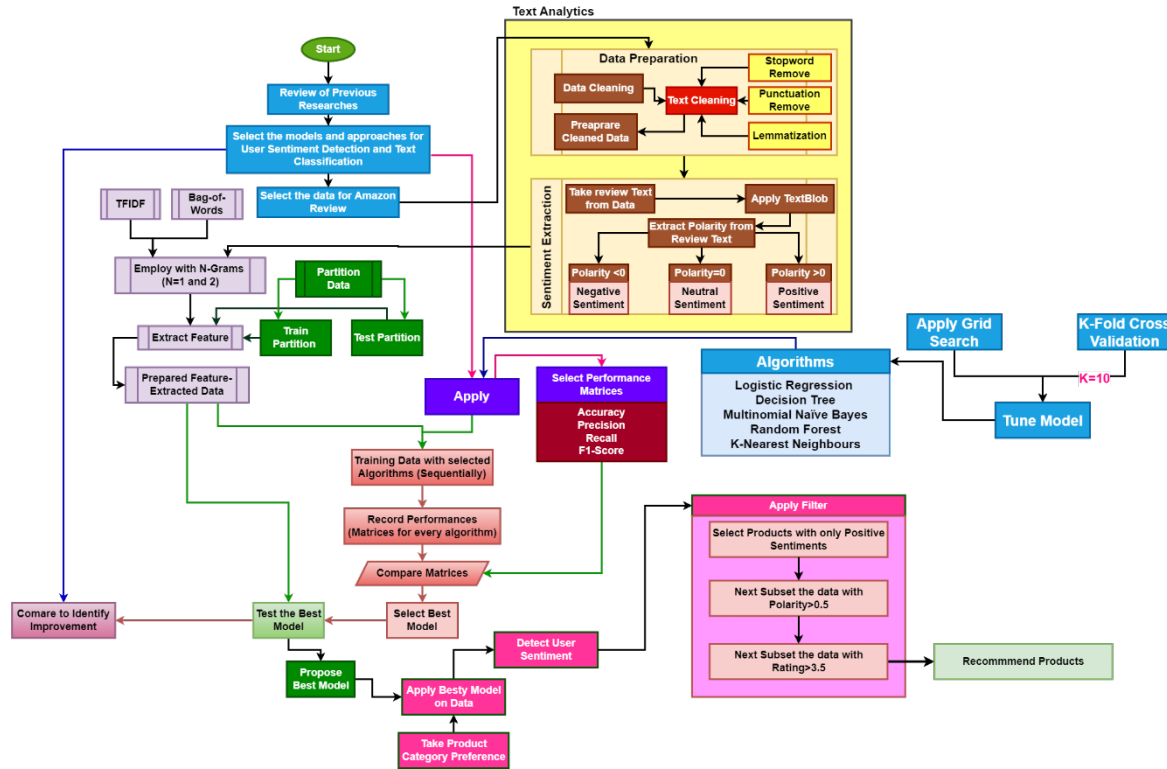


Figure 3 Proposed Methodology for Product Recommendation

4.3 TOOL TO BE USED

Python will be used for sentiment analysis and recommending Amazon products. The reasons are described below:

1. Python has rich library support for natural language processing, making it a popular choice for sentiment analysis. With its extensive libraries such as NLTK and spaCy, Python allows for efficient text processing and analysis of sentiment from textual data. Moreover, Python's versatility allows for the integration of machine learning algorithms to improve the accuracy of sentiment analysis models.
2. In addition, Python has rich library support for machine learning, which is crucial for building recommendation systems for Amazon products. By employing machine learning techniques such as collaborative filtering and content-based filtering, Python can be utilized to personalize product recommendations for users based on their preferences and behaviour.

3. Furthermore, Python provides extensive support in data and text analytics and visualization. With libraries like Pandas, NumPy, and Matplotlib, Python enables the processing, analysis, and visualization of large datasets, including textual data. This makes it suitable for extracting insights from Amazon product reviews and other textual sources to enhance the recommendation system.
4. So, Python's capabilities in natural language processing, machine learning, and data analytics make it an ideal choice for developing a sophisticated sentiment analysis and product recommendation system for Amazon. By leveraging Python's rich libraries and robust ecosystem, businesses can gain valuable insights from textual data and deliver personalized product recommendations to their customers efficiently.

4.4 CHOICE OF ALGORITHMS

The algorithms for text feature extraction and classification for review sentiment detection have been chosen by taking ideas from the literature review. The chosen algorithms have been described in this section.

4.4.1 Text Feature Extraction

Term Frequency Inverse Document Frequency

TF-IDF, which stands for Term Frequency-Inverse Document Frequency, measures the importance of a term within a document relative to a collection of documents. By assigning weights to terms based on their frequency and rarity across the corpus, TF-IDF helps in identifying keywords that are significant in a specific document.

This algorithm has been used by the following researchers:

1. Shah & Patel (2016)
2. Dzisevič (2019)
3. Du & Hu (2022)
4. Wang et al. (2012)
5. AlQahtani (2021)

Bag of Words

On the other hand, the Bag of Words approach involves representing text data as a collection of words without considering the order in which they appear. This method creates a sparse matrix where each row corresponds to a document and each column represents a unique word

in the entire corpus. The frequency of each word in a document is then used to construct the vector representation.

This algorithm has been used by the following researchers:

1. Du & Hu (2022)
2. Wang et al. (2012)
3. Sangeetha & Kumaran (2023)
4. Sharma et al. (2023)
5. Zhang et al. (2023)

4.4.2 Sentiment Classification

The following classifiers have been chosen for sentiment classification:

Table 3 Algorithms for Sentiment Classification

Algorithm	Type	Previously Applied By
Logistic Regression	Linear Model	Monsoor et al. (2023) Kumar et al. (2021) Ben et al. (2023)
Decision Tree	Tree Model	Sultan (2022) Kumar et al. (2021)
Multinomial Naïve Bayes	Naïve Bayes Model	Sultan (2022) Rathor et al. (2018) Kumar et al. (2021) AlQahtani (2021)
Random Forest	Ensemble Model	Zhao et al. (2021) Rathor et al. (2018) Kumar et al. (2021)
K-Nearest Neighbours	Neighbours Model	Sultan (2022) Monsoor et al. (2023)

Chapter-5 IMPLEMENTATION AND RESULT

5.1 DATA READING AND PROCESSING

5.1.1 Data Reading

The selected data has been obtained in a zip file from which the CSV (data file) has been read and the data is shown below:

id	dateAdded	dateUpdated	name	asins	brand	categories	primaryCategory
AVpgNzjwLJeJML43Kpxn	2015-10-30T08:59:32Z	2019-04-25T09:08:16Z	AmazonBasics AAA Performance Alkaline Batterie...	B00QWO9P0O,B00LH3DMUO	Amazonbasics	AA,AAA,Health,Electronics,Health & Household,C...	Health & Bea
AVpgNzjwLJeJML43Kpxn	2015-10-30T08:59:32Z	2019-04-25T09:08:16Z	AmazonBasics AAA Performance Alkaline Batterie...	B00QWO9P0O,B00LH3DMUO	Amazonbasics	AA,AAA,Health,Electronics,Health & Household,C...	Health & Bea
AVpgNzjwLJeJML43Kpxn	2015-10-30T08:59:32Z	2019-04-25T09:08:16Z	AmazonBasics AAA Performance Alkaline Batterie...	B00QWO9P0O,B00LH3DMUO	Amazonbasics	AA,AAA,Health,Electronics,Health & Household,C...	Health & Bea

Figure 4 Amazon Review Data

5.1.2 Investigation on Features

An investigation has been done on the data features to identify the missing values, data types etc. The result of the investigation has been presented below by displaying the data information:

```
RangeIndex: 28332 entries, 0 to 28331
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     28332 non-null  object
1   dateAdded                             28332 non-null  object
2   dateUpdated                           28332 non-null  object
3   name                                   28332 non-null  object
4   asins                                 28332 non-null  object
5   brand                                 28332 non-null  object
6   categories                             28332 non-null  object
7   primaryCategories                     28332 non-null  object
8   imageURLs                             28332 non-null  object
9   keys                                   28332 non-null  object
10  manufacturer                           28332 non-null  object
11  manufacturerNumber                     28332 non-null  object
12  reviews.date                           28332 non-null  object
13  reviews.dateSeen                       28332 non-null  object
14  reviews.didPurchase                    9 non-null      object
15  reviews.doRecommend                   16086 non-null  object
16  reviews.id                             41 non-null     float64
17  reviews.numHelpful                     16115 non-null  float64
18  reviews.rating                         28332 non-null  int64
19  reviews.sourceURLs                     28332 non-null  object
20  reviews.text                           28332 non-null  object
21  reviews.title                          28332 non-null  object
22  reviews.username                       28332 non-null  object
23  sourceURLs                             28332 non-null  object
```

Figure 5 Feature Investigation

5.2 DATA VISUALIZATION

The features of the data, such as rating, top viewers, top brands, and category of products, have been analyzed to derive meaningful insights. The outcomes of this analysis have been presented in bar charts for better visualization and comprehension.

- **Rating Analysis:** Understanding the distribution of ratings helps in identifying overall user satisfaction.
- **Top Viewers:** Analyzing the top viewers provides insights into the most active users and their sentiment trends.
- **Top Brands:** Examining the top brands reveals which brands are most frequently reviewed and the sentiment associated with them.
- **Category of Products:** Analyzing product categories helps in understanding which categories are most popular and how users feel about them.

The analyses have been done on some features and the outcomes are presented below:

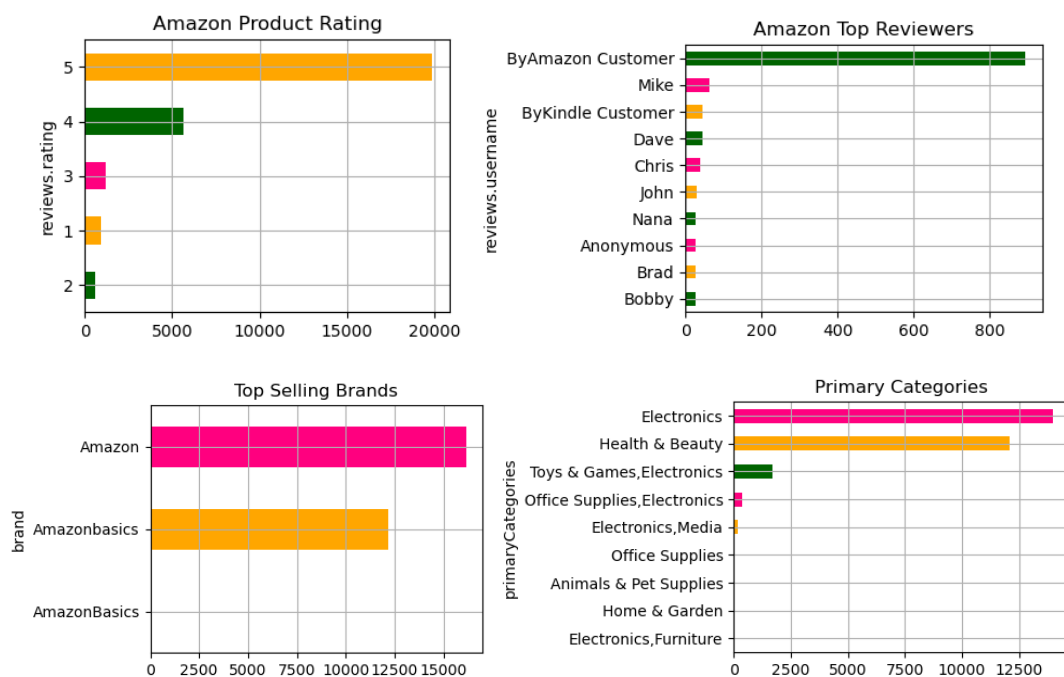


Figure 6 Visualization of Data Features

5.3 TEXT PREPROCESSING

5.3.1 Missing Value Inspection and Data Cleaning

Missing values have been identified in the data features which need to be removed in order to execute data analytics and classification of review sentiments. The missing values have been shown below before and after removing those:

Missing Values Available in Data		Cleaned	Missing Values Status after Cleaning	
Feature	Count		Feature	Count
id	0		id	0
dateAdded	0		dateAdded	0
dateUpdated	0		dateUpdated	0
name	0		name	0
asins	0		asins	0
brand	0		brand	0
categories	0		categories	0
primaryCategories	0		primaryCategories	0
imageURLs	0		imageURLs	0
keys	0		keys	0
manufacturer	0		manufacturer	0
manufacturerNumber	0		manufacturerNumber	0
reviews.date	0		reviews.date	0
reviews.dateSeen	0		reviews.dateSeen	0
reviews.didPurchase	28323		reviews.didPurchase	0
reviews.doRecommend	12246		reviews.doRecommend	0
reviews.id	28291		reviews.id	0
reviews.numHelpful	12217		reviews.numHelpful	0
reviews.rating	0		reviews.rating	0
reviews.sourceURLs	0		reviews.sourceURLs	0
reviews.text	0		reviews.text	0
reviews.title	0		reviews.title	0
reviews.username	0		reviews.username	0
sourceURLs	0		sourceURLs	0

Figure 7 Cleaning Missing Values from Features

5.3.2 Feature Merging

In the data, there are two features which contain important information on user reviews which are “reviews.text” and “reviews.title”. The title and text of the reviews have been merged to form a singleton review feature as shown below:

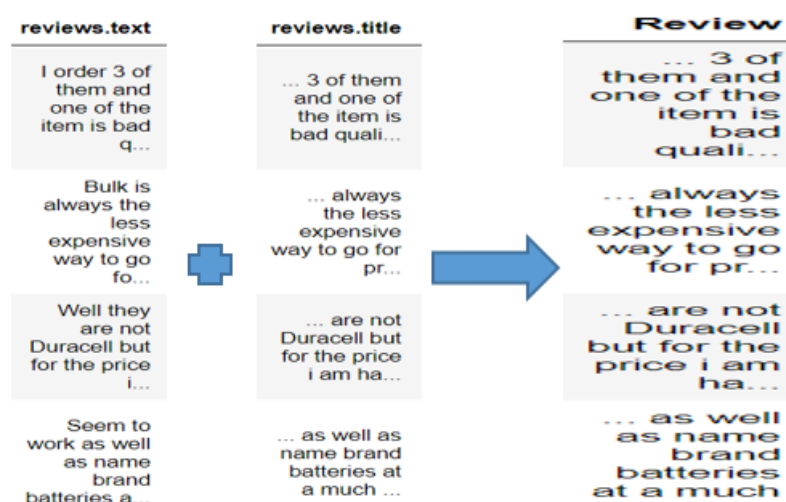


Figure 8 Merging Title and Text of Review

5.3.3 Text Cleaning

The review text contains several noises which may lead to incorrect sentiment determination, thereby falsifying the classification of sentiment. To overcome this issue, NLP techniques have been meticulously applied to clean the review text. The cleaning process involves:

- Removal of Punctuation: Eliminating any punctuation marks to avoid misinterpretation of text.
- Removal of Stopwords: Filtering out common words that do not contribute to the sentiment, such as 'is', 'and', 'the'.
- Tokenization: Breaking down the text into individual words or tokens to facilitate analysis.
- Lemmatization: Converting words to their base or root form to ensure uniformity and consistency.

The cleaned review text has been assigned to a new feature namely “Cleaned_Review” as shown below:

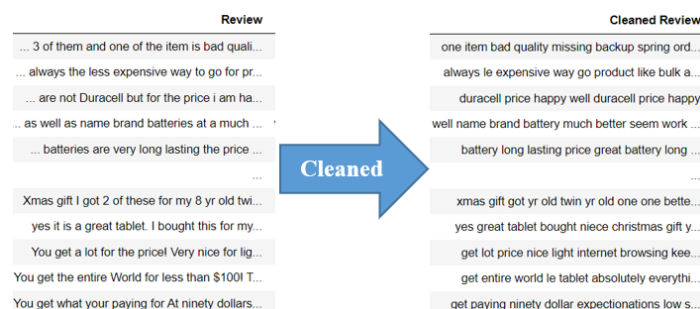


Figure 9 Cleaning Review Text

5.3.4 Sentiment Preparation

The data does not have any proper target feature containing the sentiments of users, without which the detection of sentiment cannot be accurately performed. To address this issue, TextBlob has been applied to extract polarity from the text. TextBlob is a robust NLP library that provides a simple API for diving into common natural language processing tasks.

- Polarity Extraction: TextBlob analyzes the text to determine the polarity, which ranges from -1 (negative) to +1 (positive) and the value zero stands for neutral sentiment. So, this polarity score is then used to categorize the sentiment as positive, negative, or neutral.

- **Sentiment Preparation:** Based on the extracted polarity, sentiments are prepared and assigned to the respective reviews, enabling a structured analysis of user opinions.

So, the sentiments have been extracted from the cleaned review text using TextBlob. Based on the polarity values, the sentiments have been prepared using the following process:

- Positive Sentiment $\rightarrow$ Polarity > 0
- Neutral Sentiment $\rightarrow$ Polarity $= 0$
- Negative Sentiment $\rightarrow$ Polarity < 0

After preparing sentiment, these have been stored in the data as follows:

Review	Review_Cleaned	Polarity	Sentiment
... 3 of them and one of the item is bad quali...	one item bad quality missing backup spring ord...	-0.450000	Negative
... always the less expensive way to go for pr...	always le expensive way go product like bulk a...	-0.500000	Negative
... are not Duracell but for the price i am ha...	duracell price happy well duracell price happy	0.800000	Positive
... as well as name brand batteries at a much ...	well name brand battery much better seem work ...	0.500000	Positive
... batteries are very long lasting the price ...	battery long lasting price great battery long ...	0.250000	Positive
. batteries for Christmas and the AmazonBasi...	battery christmas amazonbasics cell good haven...	0.466667	Positive
... batteries have ordered them in the past be...	battery ordered past pleased ive problame batt...	0.125000	Positive
... batteries that last quite a while then the...	battery last quite perfect nothing say well lo...	0.480000	Positive
... do not hold the amount of high power juice...	hold amount high power juice like energizer du...	0.051111	Positive
. done well by me appear to have a good shel...	done well appear good shelf life amazonbasics ...	0.300000	Positive

Figure 10 Review Sentiment Preparation

5.4 PREPARATION OF DATA

5.4.1 Partitioning Data

The dataset has been partitioned to create a training set with 80% of the data and a test set with 20% of the data. The training set will be used to train the models, and the test set will be used to predict the sentiment and evaluate model performances. This partitioning ensures that the models are trained on a substantial amount of data while maintaining a separate dataset for unbiased evaluation. The integrity of the testing process is crucial for assessing the generalizability of the models. The instances present in the classes for both partitions are listed below:

Table 4 Data Partitioning

Sentiment	Data Partition	
	Training (80%)	Testing (20%)
Positive	20171	5027
Negative	1289	324
Neutral	1205	316

5.4.2 Feature Extraction

Machine learning models cannot accept textual data directly, and the review features of the data are textual. To make the data compatible with machine learning models, feature extraction has been performed on the review (Cleaned_Review) feature using Term Frequency-Inverse Document Frequency (TFIDF) and Bag-of-Words methods with Unigram and Bigram. These techniques transform the text into numerical vectors that can be processed by machine learning algorithms. TFIDF captures the importance of terms within the context of the entire dataset, while Bag-of-Words represents the frequency of terms. Using both Unigram and Bigram allows the models to capture single words and pairs of consecutive words, providing a richer representation of the text. The number of extracted features with both two methods is presented below with the methods applied.

Table 5 Feature Extraction Processes

Process	Method	Extracted Features	
		Unigram	Bigram
Term Frequency-Inverse Document Frequency	Tfidfvectorizer	8955	95559
Bag of Words	CountVectorizer	8955	95559

5.5 SENTIMENT DETECTION RESULT

5.5.1 Assigning Classifiers

The classifiers have been assigned to a list, enabling the sequential execution of training and testing to obtain the evaluation metrics for all models. This approach ensures systematic processing and comparison of various machine learning algorithms. The classifiers include popular models such as Logistic Regression, Decision Tree, Random Forest, Naive Bayes etc.

Each classifier will be trained on the training set and evaluated on the test set to determine their performance metrics, such as accuracy, precision, recall, and F1-score.

The outlook of the classifiers is shown below:

Table 6 Outlook of Classifiers

Algorithm (Classifier)	Model Family	Outlook
Logistic Regression	Linear Model	▼ LogisticRegression LogisticRegression()
Decision Tree	Tree	▼ DecisionTreeClassifier DecisionTreeClassifier()
Multinomial Naïve Bayes	Naïve Bayes	▼ MultinomialNB MultinomialNB()
Random Forest	Ensemble	▼ RandomForestClassifier RandomForestClassifier()
K-Nearest Neighbours	Neighbours	▼ KNeighborsClassifier KNeighborsClassifier()

5.5.2 Result of Detection

The results of sentiments for Amazon product reviews have been obtained using TFIDF and Bag-of-Words methods with Unigram and Bigram. Those results are shown below. The evaluation metrics for each model provide insights into their effectiveness in predicting sentiment. For instance, accuracy indicates the proportion of correctly classified reviews, while precision and recall provide deeper insights into the performance regarding positive and negative sentiments. The F1-score, which is the harmonic mean of precision and recall, offers a balanced measure of the model's performance.

5.5.2.1 Result for TFIDF

The results of sentiment detection using TFIDF with Unigram and Bigram are shown below:

Table 7 Results Using TFIDF

Algorithm	N-Gram	Accuracy	Precision	Recall	F1-Score
Random Forest	Unigram	96.29	96.28	96.29	96.05
	Bigram	92.27	94.19	92.27	92.75
Logistic Regression	Unigram	95.94	95.85	95.94	95.63
	Bigram	91.76	91.44	91.76	89.78
Decision Tree	Unigram	95.71	95.73	95.71	95.7
	Bigram	90.82	93.07	90.82	91.6
Multinomial Naïve Bayes	Unigram	88.72	84.42	88.72	83.44
	Bigram	88.99	88.56	88.99	84.17
K-Nearest Neighbours	Unigram	43.76	90.01	43.76	55.31
	Bigram	19.52	92.52	19.52	25.17

5.5.2.2 Result for Bag-of-Words

The results of sentiment detection using Bag-of-Words with Unigram and Bigram are shown below:

Table 8 Results Using Bag-of-Words

Algorithm	N-Gram	Accuracy	Precision	Recall	F1-Score
Random Forest	Unigram	96.45	96.49	96.45	96.32
	Bigram	91.41	94.25	91.41	92.23
Logistic Regression	Unigram	97.76	97.76	97.76	97.74
	Bigram	94.39	94.03	94.39	93.92
Decision Tree	Unigram	96.82	96.83	96.82	96.8
	Bigram	90.61	93	90.61	91.37
Multinomial Naïve Bayes	Unigram	90.52	90.04	90.52	89.16
	Bigram	91.48	90.31	91.48	89.92
K-Nearest Neighbours	Unigram	90.56	93.03	90.56	91.13
	Bigram	47.91	91.99	47.91	58.91

5.5.3 Optimum Model and Final Result

5.5.3.1 Optimum Model

From the results obtained from TFIDF and Bag-of-Words, it can be seen that Logistic Regression has detected review sentiments with optimum accuracy (97.76%) with Bag-of-Words and Unigram. This performance outperformed the other models with other Vectorization processes. So, Logistic Regression has been taken as the optimum model. In addition to the optimum model, the framework which has been taken optimally is the

combination of Bag-of-Words with Logistic Regression for which 97.76% accuracy has been achieved. The optimum framework is shown below:

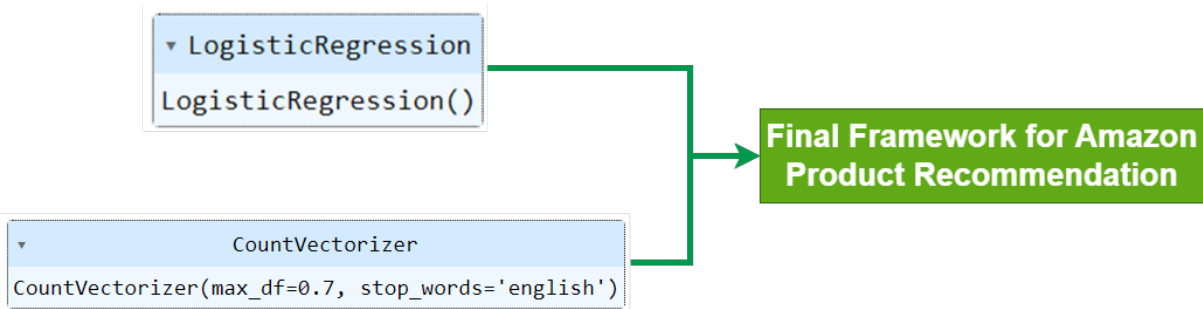


Figure 11 Optimum Model and Framework

5.5.3.2 Result

The results of the optimum framework are shown below through the confusion matrix and classification report:

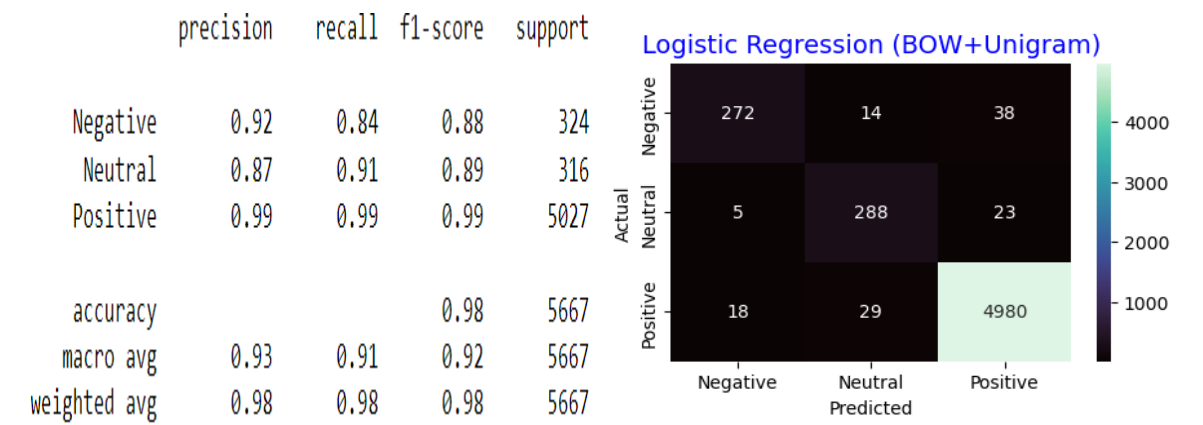


Figure 12 Results of Optimum Framework

From the above-presented results, the following outcomes can be presented:

1. The framework has classified 4980 Positive instances accurately out of 5027 test instances.
2. The framework has classified 272 Negative instances accurately out of 324 test instances.
3. The framework has classified 288 Neutral instances accurately out of 316 test instances.

So, the total classifications occurred by the framework is 127 out of 5667 total test instances with 5540 correct detection of sentiments.

5.6 AMAZON PRODUCT RECOMMENDATION

5.6.1 Data Preparation for Recommendation

5.6.1.1 Preparing Data

The data has been prepared using the optimum model by detecting sentiments on the complete data for recommendation. The predicted sentiment feature will be used as a key feature for recommendation. The recommendation data is shown below:

primaryCategories	imageURLs	reviews.rating	reviews.sourceURLs	Polarity	Sentiment	Predicted Sentiment
Health & Beauty	https://images-na.ssl-images-amazon.com/images...	3	https://www.amazon.com/product-reviews/B00QW09...	-0.450000	Negative	Negative
Health & Beauty	https://images-na.ssl-images-amazon.com/images...	4	https://www.amazon.com/product-reviews/B00QW09...	-0.500000	Negative	Negative
Health & Beauty	https://images-na.ssl-images-amazon.com/images...	5	https://www.amazon.com/product-reviews/B00QW09...	0.800000	Positive	Positive
Health & Beauty	https://images-na.ssl-images-amazon.com/images...	5	https://www.amazon.com/product-reviews/B00QW09...	0.500000	Positive	Positive

Figure 13 Recommendation Data with Predicted Sentiments

5.6.1.2 User Input for Product Category

The user input has been taken based on the primary category of the products and the outcome is shown below:

```
-----  
-----Amazon Product Recommendations-----  
-----  
Enter Keyword (For Mutiple Keywords, use Comma): Health & Beauty  
Total 12071 Filtered for the Product Type:  
  
Health & Beauty
```

Figure 14 User Input for Primary Category

It can be seen that with the user choice for Health and Beauty products, 12071 products have been initially filtered.

5.6.2 Data Filtering for Recommendation

5.6.2.1 Result of Filter-1

The first filter has been applied to select the product records with only positive predicted sentiments. This filter results in selecting 9538 products as shown below:

Total Data Selected Using Filter-1: 9538

Figure 15 Product Records Selected through Filter 1

5.6.2.2 Result of Filter-2

The second filter has been applied to select the product records with a polarity value above 0.8. This filter results in selecting 491 products as shown below:

Total Data Selected Using Filter-2: 491

Figure 16 Product Records Selected through Filter 2

5.6.2.3 Result of Filter-3

The third filter has been applied to select the product records with review rating values higher than 4.2. This filter results in selecting 471 products as shown below:

Total Data Selected Using Filter-3: 471

Figure 17 Product Records Selected through Filter 3

5.6.3 Final Amazon Product Recommendation

Finally, the recommendation of suitable Amazon products has been done with the best five products taken from Filter 3 results by sorting them using rating values. The recommended product will include the necessary details such as category, polarity, rating etc. The recommended Health and Beauty products are shown below:

Table 9 Recommendation

Sl. No.	Primary Category	Product Category	Polarity	Rating	Product Link
#1	Health & Beauty	Health Electronics	1.0	5	Link
# 2	Health & Beauty	Health Electronics	0.867	5	Link
# 3	Health & Beauty	Health Electronics	1.0	5	Link
# 4	Health & Beauty	Health Electronics	1.0	5	Link
# 5	Health & Beauty	Health Electronics	1.0	5	Link

Chapter-6 DISCUSSION AND CONCLUSION

6.1 DISCUSSION

6.1.1 Evaluation of Performance of Research

The performance of the optimum framework (Logistic Regression with BOW) has been compared with the existing research to determine the superiority of the present research. It has been seen that the present research has outperformed the existing research in detecting the sentiment of the product review and recommendations. The comparison is shown below:

Table 10 Comparison With Existing Research

Researchers	Research on	Feature Extraction Method	Text Classification Method	Domain of Research	Sentiment Detection Accuracy
Akter et al. (2021)	Sentiment Analytics	TFIDF	KNN	Machine Learning	96.25
Imaduddin et al. (2019)	Sentiment Analytics	CBOW	LSTM	Deep Learning	95.52
Huiliang et al. (2021)	Sentiment Analytics	BOW	LSIBA-ENN	Deep Learning	87.79
Sangeetha & Kumaran (2023)	Sentiment Analytics	BOW	LSTM	Deep Learning	95.4
Mehul et al. (2023)	Sentiment Analytics	TFIDF	SVM	Machine Learning	85
Alzahrani et al. (2022)	Sentiment Analytics	Word2Vec	CNN-LSTM	Deep Learning	91
Alroobaea (2022)	Sentiment Analytics with Product Recommendation	Word2Vec	RNN	Deep Learning	85
Rahman et al. (2023)	Sentiment Analytics with Product Recommendation	BOW	LSTM	Deep Learning	85
Present Research	Sentiment Analytics with Product Recommendation	BOW	Logistic Regression	Machine Learning	97.76

The comparison shows that the preset research has the superior approach in detecting product review sentiment and recommendation. The comparison is visualized below:

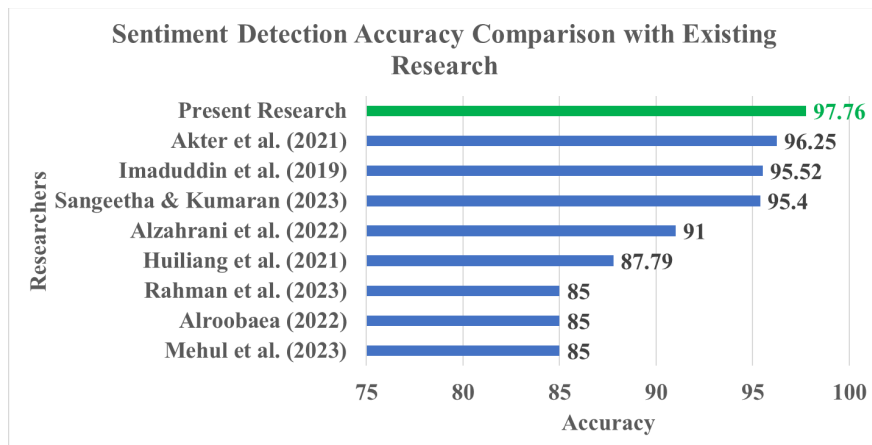


Figure 18 Sentiment Detection Accuracy Comparison with Existing Research

6.1.2 New Implications

The research has been successfully conducted for the detection of user review sentiments on products and recommending them with suitable online products. To identify the methodologies, algorithms etc. the previous research has been reviewed and certain gaps have been identified. Those gaps have been presented in [Table 1](#). In those research, the following applications were missing:

1. From the gap analysis, it has been seen that some research is related to sentiment analytics as well as product recommendations and those are Alroobaea (2022), Rahman et al. (2023) etc. Most of the research either has been conducted for sentiment analytics or product recommendations.
2. The researchers enlisted in Table 1 have employed a single feature extraction method for text vectorization. They have not used the variations of n-grams to get reliable data for sentiment classifications. As a result, the accuracies of sentiment detection were less in those research. In the present research, as multiple feature extraction methods have been used with the variations of n-gram ranges, a wider research has been established and optimum accuracy has been achieved.

6.1.3 Discussion on Research Questions

RQ 1

Bag-of-Words with Unigram has been identified as the most effective vectorization model, achieving a sentiment detection accuracy of 97.76% when used with Logistic Regression. This high level of accuracy underscores the model's capability to discern sentiment nuances within textual data. The simplicity of the Unigram approach, which considers each word independently, combined with the robustness of the Bag-of-Words representation, has proven to be a powerful combination in the sentiment analysis domain.

RQ 2

A comprehensive evaluation was conducted involving five distinct machine learning models to assess their efficacy in detecting sentiments from Amazon product reviews. These models included Naive Bayes, Support Vector Machines (SVM), Random Forest, Decision Trees, and Logistic Regression. After rigorous testing, it was observed that Logistic Regression coupled with the Bag-of-Words and Unigram model significantly outperformed the other models, achieving an impressive accuracy rate of 97.76%. This suggests that Logistic Regression, a model known for its interpretability and efficiency, is particularly well-suited for sentiment analysis tasks when applied with the right vectorization technique.

RQ 3

The user preference data has been meticulously collected and utilized to enhance the product recommendation system for Amazon. By integrating the optimized framework of Logistic Regression with Bag-of-Words and Unigram, the system can accurately identify and

recommend the most suitable products based on individual user preferences. This integration ensures that the product recommendations are not only relevant but also aligned with the detected sentiments from user reviews. Consequently, users receive highly personalized recommendations, which are likely to enhance their overall shopping experience.

6.1.4 Discussion and Fulfilment of Research Gaps

6.1.4.1 Gap-1: Feature Extraction Methods and N-grams

Feature Extraction Methods and N-grams: The initial gap identified in previous research is the limited application of diverse feature extraction methods combined with multiple n-grams. Traditional studies have often relied on singular methods, which can restrict the model's ability to capture nuanced patterns in the data. In our research, this gap has been addressed by integrating TFIDF and Bag of Words techniques with both Unigram and Bigram models. This dual approach allows for a richer and more granular analysis of textual data, enhancing the model's capability to understand and predict user preferences more accurately.

6.1.4.2 Gap-2: Computational Efficiency and Cost

Another significant gap found in existing research relates to the high computational power and cost associated with high-level architecture models used in product recommendation systems. Most researchers have employed complex neural networks and deep learning architectures, which, while effective, are resource-intensive and not feasible for all applications. Our research addresses this gap by employing a streamlined machine-learning model that achieves a remarkable 97.76% accuracy. This model leverages optimized algorithms and efficient data processing techniques, ensuring robust performance without the need for extensive computational resources.

6.1.5 Discussion and Mitigation of Challenges

6.1.5.1 Challenge 1: Handling Noisy Data with Missing Values **Challenge**

One of the primary challenges faced during data preprocessing is the presence of noisy data and missing values. Noisy data, characterized by random errors or outliers, can distort the analysis and lead to inaccurate machine-learning models. Missing values, on the other hand, pose a significant problem as most machine learning algorithms require complete datasets for effective training and prediction.

Mitigation Strategy

To address this issue, a comprehensive data-cleaning process was undertaken. This involved:

- **Data Imputation:** Techniques such as mean, median, or mode imputation were employed to fill in the missing values, ensuring that the dataset remained comprehensive.

6.1.5.2 Challenge 2: Addressing Text Noise in Sentiment Analysis

Challenge

The second significant challenge was dealing with text noise in user reviews. Text noise, including slang, abbreviations, typos, and irrelevant information, can lead to incorrect sentiment analysis. This, in turn, affects the classification of sentiments, potentially yielding falsified results.

Mitigation Strategy

Natural Language Processing (NLP) techniques were applied to clean the review text effectively. The steps included:

- **Tokenization:** Splitting the text into individual words or phrases to analyze them separately.
- **Stop Word Removal:** Filtering out common but insignificant words (e.g., "and," "the," "is") that do not contribute to the sentiment.
- **Lemmatization and Stemming:** Converting words to their base or root form to standardize the text.
- **Noise Removal:** Eliminating special characters, numbers, and other non-textual elements.

6.1.5.3 Challenge 3: Unavailability of Target Feature

Challenge

The final challenge was the unavailability of a proper target feature indicating the sentiments of user reviews. Without a predefined sentiment label, it becomes challenging to train a sentiment analysis model.

Mitigation Strategy

To overcome this issue, TextBlob, a popular NLP library, was utilized to extract polarity scores from the text. The steps involved:

- **Polarity Extraction:** TextBlob analyzes the text and assigns a polarity score ranging from -1 (negative) to +1 (positive), indicating the sentiment.
- **Sentiment Labeling:** Based on the polarity scores, the text was categorized into positive, negative, or neutral sentiments.

6.2 CONCLUSION

The present research has commenced to detect user review sentiments along with recommending Amazon products. This study aims to leverage sentiment analysis techniques to understand customer opinions and improve product recommendations on the Amazon platform. By accurately identifying the sentiments expressed in user reviews, businesses can gain valuable insights into customer satisfaction and preferences, ultimately enhancing the shopping experience. The datasets have been collected from Kaggle and text analytics have been performed to remove missing values, eliminate text noises, etc. Advanced preprocessing techniques such as tokenization, stemming, and lemmatization have been applied to ensure the data is clean and ready for analysis. Additionally, stopwords have been removed to reduce noise and improve the quality of the sentiment analysis. This meticulous data-cleaning process is crucial for obtaining reliable and accurate results in subsequent analyses. The data does not have any proper target feature containing the sentiments of users without which the detection of sentiment cannot be done. TextBlob has been applied to extract polarity based on which the sentiment has been prepared. TextBlob, a powerful natural language processing library, has been utilized to compute the sentiment polarity of each review. Polarity scores range from -1 (negative sentiment) to +1 (positive sentiment), with neutral sentiments falling around 0. These scores have been used to categorize the reviews into positive, negative, and neutral sentiments, forming the basis for further analysis and model training.

The dataset has been partitioned to create a training set with 80% of the data and a test set with 20% of the data. The training set will be used to train the models and the test set will be used to predict the sentiment and evaluate model performances. This split ensures that the models are trained on a substantial amount of data while still leaving a significant portion for validation. Various machine learning algorithms have been employed to build robust sentiment analysis models.

Machine learning model cannot accept textual data and the review features of the data is textual. To make the data compatible to machine learning models, feature extraction has been done on review (Cleaned_Review) feature using TFIDF and Bag-of-Words methods with Unigram and Bigram. This process involves converting the textual data into numerical vectors that can be processed by the machine learning algorithms. The Term Frequency-Inverse Document Frequency (TFIDF) method helps in understanding the importance of a word in a document relative to a collection of documents, whereas the Bag-of-Words method focuses on the frequency of words in the text. These techniques ensure that the textual data is effectively transformed into a structured format suitable for further analysis. The classifiers have been assigned to a list so that the sequential execution of training and testing can be done following to obtain the evaluation metrics for all models. This approach allows for systematic and efficient model evaluation. By iterating through the list of classifiers, each model can be trained on the training data and subsequently test it on the validation data. This sequential process helps in comparing the performance metrics such as accuracy, precision, recall, and F1 score across different models, thereby aiding in the selection of the most effective classifier for the task at hand.

The result of sentiment detection has been achieved optimally using Logistic Regression with Bag-of-Words and Unigram with an accuracy rate of 97.76%. Logistic Regression, a robust and widely used statistical model, has demonstrated exceptional performance in sentiment analysis. By employing the Bag-of-Words approach combined with Unigram features, the model effectively captures the sentiment conveyed in the reviews. The high accuracy rate signifies the model's proficiency in distinguishing between positive and negative sentiments, making it a reliable tool for sentiment detection in textual data. The optimum framework has been employed to recommend products to the users by taking user preference and the top five products have been recommended to the users. This recommendation system leverages the insights gained from user preferences and sentiment analysis to suggest products that align with the users' interests. By analyzing past behaviour, purchase history, and sentiment expressed in reviews, the system can generate personalized recommendations. This enhances the user experience by providing tailored product suggestions, thereby increasing user satisfaction and engagement.

6.2.1 Present Limitations

The current project and research are subject to the following limitations:

1. The dataset utilized for the research was sourced from a single origin.
2. The selected dataset encompasses a limited number of product records.
3. The algorithms employed are restricted to a few machine-learning techniques.
4. No deep-learning algorithms have been applied.
5. Sentiment analysis was conducted solely using polarity values, without considering subjectivity values.

6.2.2 Future Recommendations

Future work could explore the application of deep learning techniques such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks for sentiment analysis. These advanced models have the potential to capture more nuanced patterns in textual data, leading to even more accurate sentiment detection. Moreover, expanding the dataset to include reviews from multiple e-commerce platforms could provide a more comprehensive understanding of user sentiments across different contexts.

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APPENDIX

Function to load the data from zip files

```
def LoadDFAM():
```

```
    flavl = os.listdir(os.getcwd()) # Getting the list of files in the current directory
```

```
    for x in flavl:
```

```
        # Finding the zip file containing data by checking specific keywords in file names
```

```
            if "Datafiniti" in x or "Consumer_Reviews" in x or "Amazon" in x:
```

```
                loaded_zip = x
```

```
        # Reading the CSV file inside the zip and storing it in 'dfam' dataframe
```

```
        dfam = pa.read_csv(loaded_zip, compression='zip')
```

```
        # Printing the number of records and features in the dataframe
```

```
        print("Records: {} \n Features: {}".format(dfam.shape[0], dfam.shape[1]))
```

```
        # Finding categorical and numerical features in the dataframe
```

```
        catf = dfam.dtypes[dfam.dtypes == 'object'].index.tolist()
```

```
        numf = dfam.dtypes[dfam.dtypes != 'object'].index.tolist()
```

```
        print("-----")
```

```
        print("Categorical Features: {} \n Numerical Features: {}".format(len(catf), len(numf)))
```

```
    return dfam
```

```
DataAmazon=LoadDFAM()
```

```
DataAmazon.head(3)
```

```
# Function to investigate and handle missing values in the dataframe
```

```
def FetInvestigate(dfam):
```

```
    # Printing information about the dataframe
```

```
    print(dfam.info())
```

Counting missing values in each column

```
print(dfam.isnull().sum())
```

```
dfammissing = sum(dfam.isnull().sum())
```

```
print("\nInitial Missing Values: {}".format(dfammissing))
```

Handling missing values: filling categorical columns with mode and numerical columns with mean

```
if dfammissing > 0:
```

```
    catf = dfam.dtypes[dfam.dtypes == 'object'].index.tolist()
```

```
    numf = dfam.dtypes[dfam.dtypes != 'object'].index.tolist()
```

```
    for m in catf:
```

```
        dfam[m] = dfam[m].fillna(dfam[m].mode()[0])
```

```
    for n in numf:
```

```
        dfam[n] = dfam[n].fillna(dfam[n].mean())
```

```
dfammissing1 = sum(dfam.isnull().sum())
```

```
print("\nFinal Missing Values: {}".format(dfammissing1))
```

```
print(dfam.isnull().sum())
```

```
return dfam
```

Loading data from Amazon and cleaning it

```
Cln_DataAmazon=FetInvestigate(DataAmazon)
```

```
Cln_DataAmazon.head()
```

Function to concatenate two text features into one

```
def ConcatFet(dfam, f1, f2):
```

```
    scrh = 'Review'
```

```
    allwed = 'object'
```

```
if dfam[f1].dtypes == allwed and dfam[f2].dtypes == allwed:
```

```
    print("Success in Merging")
```

```
    dfreview = Cln_DataAmazon[f1].tolist()
```

```
    dftitle = Cln_DataAmazon[f2].tolist()
```

```
    review_fin_fet = []
```

```
    for x in range(len(dfreview)):
```

```
        review_fin_fet.append(dftitle[x]+' '+dfreview[x])
```

```
    if scrh in dfam.columns.tolist():
```

```
        dfam = dfam.drop('Review', axis=1)
```

```
    dfam[scrh] = review_fin_fet
```

```
    return dfam
```

```
else:
```

```
    print("Error in Merging")
```

```
    return dfam
```

```
# Concatenating text features 'reviews.text' and 'reviews.title'
```

```
f1='reviews.text'
```

```
f2='reviews.title'
```

```
Cln_DataAmazon_1=ConcatFet(Cln_DataAmazon,f1,f2)
```

```
Cln_DataAmazon_1.head()
```

```
# Function to clean text data in the 'Review' column
```

```
def Clean_Review_Text(DfAmaz):
```

```
    def LemText(txdthon):
```

```
        return [Imma.lemmatize(t) for t in txdthon]
```

```

def Clean_Row_Text(dfamrw):

    nltkstp = list(nltk.corpus.stopwords.words('english'))

    txt1 = ""

    wrdtodel = ["rt", "RT", "http", "HTTP", "https", "HTTPS", "com", "COM"]

    for d in dfamrw:

        if d.isalpha() == True:

            txt1 += d

        if d.isalnum() == False:

            pass

        if d == " " or d == "." or d == "," or d == ";":

            txt1 += " "

    tkreview = str(txt1.lower()).split()

    tk2 = []

    for i in tkreview:

        if i not in wrdtodel:

            tk2.append(i)

    tk3 = []

    for i in tk2:

        if i not in nltkstp:

            tk3.append(i)

    tkjn = ''.join(tk3)

    tk3 = tkjn.split()

    tk4 = LemText(tk3)

    return ''.join(tk4)

```

```

RevTextClean = []

TxtReview = DfAmaz["Review"].tolist()

for i in TxtReview:

    RevTextClean.append(Clean_Row_Text(i))

DfAmaz['Review_Cleaned'] = RevTextClean

return DfAmaz

```

Cleaning the 'Review' column text data

```

TxClN_DataAmazon = Clean_Review_Text(Cln_DataAmazon_1)

TxClN_DataAmazon.head(3)

```

Extracting relevant columns for analysis

```

TxClN_DataAmazon1 = TxClN_DataAmazon[['Review', 'Review_Cleaned']]

```

Plotting graphs for analysis

```

knd = 'barh'

fets = ['primaryCategories', 'brand', 'reviews.username', 'reviews.rating']

grphtml = ["Primary Categories", "Top Selling Brands", "Amazon Top Reviewers", 'Amazon
Product Rating']

fsz = (4, 3)

clsq = ["#006400", "#FFA600", "#FF007F"]

for f in range(len(fets)):

    TxClN_DataAmazon[fets[f]].value_counts().iloc[:10][::-1].plot(kind=knd, title=grphtml[f],
    figsize=fsz, color=clsq)

dfa_vs.grid()

```

```

dfa_vs.show()

TxCln_DataAmazon1=TxCln_DataAmazon[['Review','Review_Cleaned']]

TxCln_DataAmazon1.head()

# Sentiment analysis on cleaned review text

ReviewTxt = TxCln_DataAmazon1['Review_Cleaned'].tolist()

polarity_review = []

sentiment_products = []

for i in ReviewTxt:

    sent_rv = TextBlob(str(i)).sentiment

    polarity_review.append(sent_rv[0])

for i in range(len(polarity_review)):

    if polarity_review[i] > 0:

        sentiment_products.append("Positive")

    elif polarity_review[i] < 0:

        sentiment_products.append("Negative")

    else:

        sentiment_products.append("Neutral")

TxCln_DataAmazon1['Polarity'] = polarity_review

TxCln_DataAmazon1['Sentiment'] = sentiment_products

TxCln_DataAmazon1 = TxCln_DataAmazon1.reset_index(drop=True)

TxCln_DataAmazon1.head(10)

# Counting the occurrences of each sentiment label in the dataset

TxCln_DataAmazon1['Sentiment'].value_counts()

# Creating a copy of the dataset for manipulation

```

```

data=TxCln_DataAmazon1.copy()

# Separating the features(X) and the target variable(Y)

X=data['Review_Cleaned']. #Features : Cleaned review text

y=data['Sentiment']    #Target : Sentiment labels

# Splitting the dataset into training and testing sets (80% training, 20% testing)

DtXTrn,DtXTst,DtyTrn,DtyTst=model_selection.train_test_split(X,y,train_size=0.8,random_
state=22)

# Displaying the distribution of sentiment labels in the test set

print(DtyTst.value_counts())

# Displaying the distribution of sentiment labels in the training set

print(DtyTrn.value_counts())

# List of models to be used for classification

MdAmz=[

    linear_model.LogisticRegression(), # Logistic Regression model

    tree.DecisionTreeClassifier(), # Decision Tree model

    naive_bayes.MultinomialNB(), # Multinomial Naive Bayes model

    ensemble.RandomForestClassifier(), # Random Forest model

    neighbors.KNeighborsClassifier() # K-Nearest Neighbors model

]

# Corresponding names of the models for easier identification in output

ClAmz=[

    "Logistic Regression ",

```

```

"Decision Tree",

"Multinomial Naïve Bayes",

"Random Forest",

"K-Nearest Neighbours"

]

# Function to apply TF-IDF vectorization to the training and test data

def Func_TFIDF(DtXTrn,DtXTst):

    TrDtTf=[].          # List to store TF-IDF vectorized training data

    TsDtTf=[]           # List to store TF-IDF vectorized testing data

    ngs_val=["Unigram","Bigram"] # Types of n-grams to be used

    # Loop to create both Unigram and Bigram TF-IDF features

    for i in range(2):

        FETFIDF=feature_extraction.text.TfidfVectorizer(stop_words='english',
        analyzer='word',max_df=0.7,ngram_range=(i+1,i+1))

        TrDtTf.append(FETFIDF.fit_transform(DtXTrn)) # Fit and transform the training data

        TsDtTf.append(FETFIDF.transform(DtXTst)) # Transform the testing data (no fitting)

        # Printing the number of features extracted for Unigram and Bigram

        print("Total    Extracted    Features    in    Vectorised    Data    for    {}    TFIDF:
        {}".format(ngs_val[i],TrDtTf[i].shape[1]))

    return TrDtTf,TsDtTf.    # Return the vectorized training and testing data


# Function to apply Bag-of-Words (BOW) vectorization to the training and test data

def Func_BOW(DtXTrn,DtXTst):

    TrDtBOW=[]          # List to store BOW vectorized training data

    TsDtBOW=[]           # List to store BOW vectorized testing data

```



```

print("\n")

ngs_val=["Unigram","Bigram"]

# Loop to create both Unigram and Bigram BOW features

for i in range(2):

    FECV=feature_extraction.text.CountVectorizer(stop_words='english',
    analyzer='word',max_df=0.7,ngram_range=(i+1,i+1))

    TrDtBOW.append(FECV.fit_transform(DtXTrn))

    TsDtBOW.append(FECV.transform(DtXTst))

    # Print the number of features extracted for Unigram and Bigram

    print("Total    Extracted    Features    in    Vectorised    Data    for    {}    CV:
    {}".format(ngs_val[i],TrDtBOW[i].shape[1]))

    return TrDtBOW,TsDtBOW

# Applying TF-IDF vectorization to the training and test data

TrDtTf,TsDtTf=Func_TFIDF(DtXTrn,DtXTst)

# Applying Bag-of-Words (BOW) vectorization to the training and test data

TrDtBOW,TsDtBOW=Func_BOW(DtXTrn,DtXTst)

# Initializing lists to store results for each model and vectorization type

ResMtTFAmz=[[],[],[],[],[],[],[],[],[],[],[],[]]

ngs_val=["Unigram","Bigram"]

mts_avg="weighted" # Scoring method for precision, recall, and F1-Score

ptvl=4 # Precision of results to 4 decimal points

# Loop through the TF-IDF vectorized data (Unigram and Bigram)

for tlp in range(len(TrDtTf)):

    print("                TFIDF + {}".format(ngs_val[tlp]))

    for i in range(len(MdAmz)):

```

```

print("      {} with {}".format(ClAmz[i],ngs_val[tlp]))

MdTf = MdAmz[i]      # Select the model from the list

MdTf.fit(TrDtTf[tlp], DtyTrn) # Fit the model to the training data

# Predicting sentiment labels for the test data

resprtf=MdTf.predict(TsDtTf[tlp])

# Calculating accuracy and store it in the results list

AccTfTst=round(metrics.accuracy_score(DtyTst,resprtf),ptvl)*100

ResMtTFAmz[tlp][0].append(AccTfTst)

#Calculating and storing precision, recall, and F1-Score

ResMtTFAmz[tlp][1].append(round(metrics.precision_score(DtyTst,resprtf,average=mts_avg
),ptvl)*100)

ResMtTFAmz[tlp][2].append(round(metrics.recall_score(DtyTst,resprtf,average=mts_avg),pt
vl)*100)

ResMtTFAmz[tlp][3].append(round(metrics.f1_score(DtyTst,resprtf,average=mts_avg),ptvl
*100)

# Creating a confusion matrix

ct=pa.crosstab(DtyTst,    resprtf,    rownames=['Actual'],    colnames=['Predicted'],
margins=True)

ResMtTFAmz[tlp][4].append(ct)

# Plotting and displaying the confusion matrix as a heatmap

dfa_vs.figure(figsize=(5,3))

dfa_vs.title("{} (TFIDF+ {})".format(ClAmz[i],ngs_val[tlp]),fontsize=14,color="b")

se.heatmap(ct.iloc[:3,:3],annot=True,cmap="mako",fmt="d")

dfa_vs.show()

```

```

p=round(metrics.precision_score(DtyTst,resprtf,average=mts_avg),ptvl)*100

r=round(metrics.recall_score(DtyTst,resprtf,average=mts_avg),ptvl)*100

f=round(metrics.f1_score(DtyTst,resprtf,average=mts_avg),ptvl)*100

# Storing the classification report

ResMtTFAmz[tlp][5].append(metrics.classification_report(DtyTst, resprtf))

# Printing the classification report and performance metrics

print(metrics.classification_report(DtyTst, resprtf))

print("Accuracy : {}%\nPrecision : {}%".format(AccTfTst,p))

print("Recall : {}%\nF1-Score : {}%".format(r,f))

# Initializing a list to store DataFrames for each n-gram model (Unigram, Bigram)

AllTFDf=[]

# Loop through the TF-IDF results and create DataFrames

for i in range(2):

    DfAmTf=pa.DataFrame({

        "Algorithm":ClAmz,

        "Accuracy":ResMtTFAmz[i][0],

        "Precision":ResMtTFAmz[i][1],

        "Recall":ResMtTFAmz[i][2],

        "F1-Score":ResMtTFAmz[i][3]

    })

    AllTFDf.append(DfAmTf) # Append the DataFrame to the list

    plt_cols=DfAmTf.columns.tolist()[1:] # List of metric names for plotting

# Loop through each metric and create bar plots

    for x in plt_cols:

```

```

DfAmTf=DfAmTf.sort_values(by=x,ascending=False)

dfa_vs.figure(figsize=(6,4))

dfa_vs.title("Comparison of {}(TFIDF+{}).format(x,ngs_val[i]),fontsize=16)


dfa_vs.barh(DfAmTf['Algorithm'],DfAmTf[x],color=["#FF4500","#FC6C85","#00FA9A",
"#E2F516","#EB5406"])

dfa_vs.xlabel("Algorithm",fontsize=13)

dfa_vs.ylabel("{}".format(x),fontsize=13)

for tfid, tfvl in enumerate(DfAmTf[x]): # Annotate the bar plot with the values

    dfa_vs.text(tfvl, tfid, str(tfvl))

dfa_vs.show()


DfAmTf.sort_values(by="Accuracy",ascending=False).to_csv("{}_TFIDF.csv".format(ngs_val[i]))
# Save the sorted DataFrame as a CSV file

# Initializing a list to store metrics for Bag-of-Words models

ResMtBOWAmz=[[],[],[],[],[],[],[],[],[],[],[],[]]

ngs_val=["Unigram","Bigram"]

mts_avg="weighted"

ptvl=4


# Loop through the Bag-of-Words data

for tlp in range(len(TrDtBOW)):

    print("Bag-of-Words + {}".format(ngs_val[tlp])) # Print the current algorithm and n-gram type

# Loop through the algorithms

for i in range(len(MdAmz)):

```

```

print("          {} with {}".format(ClAmz[i],ngs_val[tlp]))

MdBOW = MdAmz[i] # Select the model

MdBOW.fit(TrDtBOW[tlp], DtyTrn) # Fit the model on the training data

resprBOW=MdBOW.predict(TsDtBOW[tlp]) # Predict using the model on the test data

# Calculating accuracy and rounding it

AccBOWTst=round(metrics.accuracy_score(DtyTst,resprBOW),ptvl)*100

ResMtBOWAmz[tlp][0].append(AccBOWTst) # Store accuracy


ResMtBOWAmz[tlp][1].append(round(metrics.precision_score(DtyTst,resprBOW,average=
mts_avg),ptvl)*100)    # Store precision


ResMtBOWAmz[tlp][2].append(round(metrics.recall_score(DtyTst,resprBOW,average=mts_
avg),ptvl)*100) # Store recall


ResMtBOWAmz[tlp][3].append(round(metrics.f1_score(DtyTst,resprBOW,average=mts_av
g),ptvl)*100) # Store F1-Score


# Creating a confusion matrix

ct=pa.crosstab(DtyTst,  resprBOW,  rownames=['Actual'],  colnames=['Predicted'],
margins=True)

ResMtBOWAmz[tlp][4].append(ct) # Store confusion matrix

dfa_vs.figure(figsize=(5,3))

dfa_vs.title("{} (BOW+{})".format(ClAmz[i],ngs_val[tlp]),fontsize=14,color="b")

se.heatmap(ct.iloc[:3,:3],annot=True,cmap="mako",fmt="d")

dfa_vs.show()


# Calculating and round precision,recall_score,f1_Score

p=round(metrics.precision_score(DtyTst,resprBOW,average=mts_avg),ptvl)*100

```

```

r=round(metrics.recall_score(DtyTst,resprBOW,average=mts_avg),ptvl)*100

f=round(metrics.f1_score(DtyTst,resprBOW,average=mts_avg),ptvl)*100

# Store and print the classification report

ResMtBOWAmz[tlp][5].append(metrics.classification_report(DtyTst, resprBOW))

print(metrics.classification_report(DtyTst, resprBOW))

print("Accuracy : {}%\nPrecision : {}%".format(AccBOWTst,p)) # Print accuracy and
precision

print("Recall : {}%\nF1-Score : {}%".format(r,f)) # Print recall and F1-Score

# Initializing a list to store DataFrames for Bag-of-Words results

AllBOWDf=[]

# Loop through the n-gram types

for i in range(2):

    DfAmBOW=pa.DataFrame({

        "Algorithm":ClAmz,

        "Accuracy":ResMtBOWAmz[i][0],

        "Precision":ResMtBOWAmz[i][1],

        "Recall":ResMtBOWAmz[i][2],

        "F1-Score":ResMtBOWAmz[i][3]

    }) # Create a DataFrame with the metrics for the current n-gram type

# Appending the DataFrame to the list

    AllBOWDf.append(DfAmBOW)

# Get the list of metric columns

    plt_cols=DfAmBOW.columns.tolist()[1:]

# Loop through each metric and create bar plots

```

```

for x in plt_cols:

    DfAmBOW=DfAmBOW.sort_values(by=x,ascending=False) # Sort the DataFrame
based on the current metric in descending order

    dfa_vs.figure(figsize=(6,4))

    dfa_vs.title("Comparison of {} (Bag-of-Words+{}).format(x,ngs_val[i]),fontsize=16)

# Creating a horizontal bar plot
    dfa_vs.barh(DfAmBOW['Algorithm'],DfAmBOW[x],color=["#FF4500","#FC6C85","#00FA
9A","#E2F516","#EB5406"])

    dfa_vs.xlabel("Algorithm",fontsize=13)

    dfa_vs.ylabel("{} ".format(x),fontsize=13)

    for BOWid, BOWvl in enumerate(DfAmBOW[x]):

        dfa_vs.text(BOWvl, BOWid, str(BOWvl))

    dfa_vs.show()


DfAmBOW.sort_values(by="Accuracy",ascending=False).to_csv("{}_BOW.csv".format(ngs
_val[i])) # Save the sorted DataFrame as a CSV file


# Initializing lists to store the best algorithms and accuracies for TFIDF
AlgTFIDF,AcrTFIDF=[],[]

# Loop through the DataFrames for TFIDF results
for x in range(len(AllTFDf)):

    AlgTFIDF.append(AllTFDf[x].sort_values(by="Accuracy",ascending=False).iloc[0][0])

    AcrTFIDF.append(AllTFDf[x].sort_values(by="Accuracy",ascending=False).iloc[0][1])

# Initializing lists to store the best algorithms and accuracies for Bag-of-Words
AlgBOW,AcrBOW=[],[]

# Looping through the DataFrames for Bag-of-Words results

```

```

for x in range(len(AllBOWDf)):

    AlgBOW.append(AllBOWDf[0].sort_values(by="Accuracy",ascending=False).iloc[0][0])

    AcrBOW.append(AllBOWDf[x].sort_values(by="Accuracy",ascending=False).iloc[0][1])

# Combining the best algorithms from both TFIDF and Bag-of-Words

AlgAll=[AlgTFIDF,AlgBOW]

# Combining the best accuracies from both TFIDF and Bag-of-Words

AcrAll=[AcrTFIDF,AcrBOW]

# Defining the vectorization techniques

vct=['TFIDF','Bag-of-Words']

# Determine the optimum model and vectorization technique

for a in range(len(AlgAll)):

    if len(nu.unique(nu.array(AlgAll[a])))!=1:

        pass

    else:

        vl=AcrAll[a].index(max(AcrAll[a]))

        OptAcr=AcrAll[a][vl]

        OptAlg=AlgAll[a][vl]

        VctOpt=vct[a]

        NgOpt=ngs_val[vl]

# Print the optimum results

print("Optimum Model: {} \nAccuracy: {} % \nOptimum Vectorization Process: {} \nOptimum
N-Gram: {}".format(OptAlg, OptAcr, VctOpt, NgOpt))

# Selecting the final model based on the optimum algorithm

FinMdl=MdAmz[C1Amz.index(OptAlg)]

```



```

print(FinMdl) # Print the final model

# Initializing the vectorizer with optimal parameters

OptVectModel=text.CountVectorizer(stop_words='english',
analyzer='word',max_df=0.7,ngram_range=(1,1))

TrComplete=OptVectModel.fit_transform(data['Review_Cleaned'])

FinMdl=FinMdl.fit(TrComplete, data['Sentiment'])

filename = 'Model_Recom.sav' # Save the model for future use

pickle.dump(FinMdl, open(filename, 'wb'))

MdlLoad = pickle.load(open(filename, 'rb')) # load the model

print(MdlLoad)

OptVectModel=text.CountVectorizer(stop_words='english',
analyzer='word',max_df=0.7,ngram_range=(1,1))

TrComplete=OptVectModel.fit_transform(data['Review_Cleaned'])

# Predicting the sentiment using the loaded model

PrdTrn=MdlLoad.predict(TrComplete)

# Adding the predicted sentiment to the dataset

try:

    data['Predicted Sentiment']=PrdTrn

except:

    pass

data

# To get user input for keywords to filter products

InputUser=input("Enter Keyword (For Mutiple Keywords, use Comma): ")

InputUserlst=InputUser.split(",")

pcat=data['primaryCategories'].tolist()

```

```

idx_keys=[]

for iu in range(len(InputUserlst)):

    for pc in range(len(pcat)):

        if InputUserlst[iu] in pcat[pc]:

            idx_keys.append(pc)

idx_keys=nu.unique(nu.array(idx_keys)).tolist()

RecomAmazon=data.iloc[idx_keys] # Filter the dataset based on the indices

print("Total      {}      Filtered      for      the      Product      Type:
\n".format(len(RecomAmazon)),*InputUserlst,sep="\n")

RecomAmazon.head() # Display the first few rows of the filtered dataset

# Filtering the dataset to include only positive sentiment

A1=RecomAmazon[RecomAmazon['Predicted Sentiment']=="Positive"]

print("Total Data Selected Using Filter-1: {}".format(len(A1)))

A1.head()

# Filtering the dataset further based on polarity

A2=A1[A1['Polarity']>0.8]

print("Total Data Selected Using Filter-2: {}".format(len(A2)))

A2.head()

# Filtering the dataset further based on rating

A3=A2[A2['reviews.rating']>4.2]

print("Total Data Selected Using Filter-3: {}".format(len(A3)))

A3.head()

# Resetting the index of the final filtered data

RecomAmazon1=A3.reset_index(drop=True)

```

```

RecomAmazon2=RecomAmazon1.sort_values(by=["reviews.rating"],ascending=False).reset_index(drop=True)

RcAmTopFive=RecomAmazon2.drop_duplicates().iloc[:5]

RcAmTopFive

# Getting the list of categories for the top 5 products

racat=RcAmTopFive['categories'].tolist()

rapc=RcAmTopFive['primaryCategories'].tolist()

raim=RcAmTopFive['imageURLs'].tolist()

rarr=RcAmTopFive['reviews.rating'].tolist()

rapl=RcAmTopFive['Polarity'].tolist()

for tp5 in range(len(racat)):

    print("\nPrimary Category: {}".format(rapc[tp5]))

    print("\nProduct Category: {}".format(' '.join(racat[tp5].split(",")[:4])))

    print("\nPolarity: {}".format(round(rapl[tp5],3)))

    print("\nRating: {}".format(rarr[tp5]))

    print("\nLinks: {}\n".format(raim[tp5]))

```